

1) How do you know if a DFA M .

a) accepts the empty string.

If the start state is also an accept state.

b recognizes the empty language.

If the DFA has no path to the accept state from the start.

2 in class $\delta^*(q, w)$ by decomposing w into $w = xa$ where $a \in \Sigma$.

also we define $\delta_L^*(q, w)$ by $w = ax$.

a write definition. $\delta_L^*(q, w) = q$ if $w = \epsilon$

$$\delta_L^*(q, w) = \delta_L^*(\delta(q, a), x) \text{ if } w = ax$$

b prove by induction

$$\forall q \in Q \quad \forall w \in \Sigma^* \text{ prove } \delta^*(q, w) = \delta_L^*(q, w)$$

$$\text{base case if } w = \epsilon, \delta^*(q, w) = q = \delta_L^*(q, w)$$

$$\text{if } w = a \text{ where } a \in \Sigma, \delta^*(q, w) = \delta(q, a) = \delta_L^*(q, w)$$

Inductive case assume $\delta^*(q, w) = \delta_L^*(q, w)$ &

$$\text{when } |w| = n+1 \quad w = axb$$

$$\begin{aligned} \delta^*(q, w) &= \delta(\delta^*(q, ax), b) \\ &= \delta(\delta_L^*(q, ax), b) \\ &= \delta(\delta_L^*(\delta(q, a), x), b) \\ &= \delta(\delta^*(\delta(q, a), x), b) \\ &= \delta^*(\delta(\delta(q, a), x), b) \\ &= \delta^*(\delta(q, ax), b) \\ &= \delta_L^*(q, w) \end{aligned}$$

c another way to decompose w .

$$\delta_{LR}^*(q, w) = \delta_{LR}^*(\delta_{LR}^*(q, x), y) \text{ if } |w| \geq 2$$

$$\delta_{LR}^*(q, w) = \delta(q, w) \text{ if } |w| = 1$$

$$\delta_{LR}^*(q, w) = q \text{ if } w = \epsilon$$

d prove by induction new definition is equivalent to $\delta^*(q, w)$

Base case if $w = \epsilon$ $\delta_{LR}^*(q, \epsilon) = q = \delta^*(q, \epsilon)$

If $w = a$ $a \in \Sigma$ $\delta_{LR}^*(q, a) = \delta(q, a) = \delta^*(q, a)$

inductive assume $\delta_{LR}^*(q, w) = \delta^*(q, w)$ for $|w| \leq n$

for $|w| = n+1$ let $w = xy$

$$\delta_{LR}^*(q, w) = \delta^*(q, w)$$

Base case if $|x| = 1$ $\delta_{LR}^*(q, xy) = \delta_L^*(q, xy) = \delta^*(q, xy)$

assume $\delta_{LR}^*(q, w) = \delta^*(q, w)$ for $|w| = n+1$ and $|x| = n$

when $|x| = n+1$ let $x = za$

$$\begin{aligned} \delta_{LR}^*(q, za y) &= \delta_{LR}^*(\delta_{LR}^*(q, za), y) \\ &= \delta^*(\delta^*(q, za), y) \\ &= \delta^*(\delta^*(q, z), ay) \\ &= \delta^*(\delta^*(q, z), ay) \\ &= \delta^*(q, za y) \end{aligned}$$

3 a prove that the 4 State and 3 State solutions to lang divide by 4 are equivalent

Pair	Final	on 0	on 1
$q_0 q_0$	Y, Y	$q_0 q_0$	$q_1 q_1$
$q_1 q_1$	n, n	$q_2 q_2$	$q_3 q_1$
$q_3 q_1$	n, n	$q_2 q_2$	$q_3 q_1$
$q_2 q_2$	n, n	$q_0 q_0$	$q_1 q_1$



pair final on 0
 $q_0 q_0$ (Y, Y) q_0 trap

string: 0

4) extend δ to δ^* write full definition

$$\begin{aligned} \delta_p^*(q, w) &= q \text{ if } w = \epsilon \\ \delta_p^*(q, xa) &= \delta_p(q', a) \text{ if } q' \text{ is defined and } \delta_p(q', a) \text{ is defined} \\ \delta_p^*(q, xa) &\text{ is undefined if } q' \text{ is undefined or } \delta_p(q', a) \text{ is undefined where} \\ & q' = \delta_p^*(q, x) \text{ and } w = xa \end{aligned}$$

5) prove $|uv| = |u| + |v|$ by induction on $|u|$

Base case if $u = \epsilon$ $|uv| = |v| = |u| + |v|$

inductive assume $|u| = n$

for $|u| = n+1$ $u = wa$

$$\begin{aligned} |uv| &= |wau| = |w| + |au| \\ &= |w| + 1 + |v| \\ &= n+1 + |v| \end{aligned}$$

b) prove $(uv)^R = v^R u^R$

Solution definition of x^R $x^R = \epsilon$ if $x = \epsilon$
 $x^R = x^R a$ if $x = ax$ where $a \in \Sigma$.

Base if $u \in \Sigma$ statement holds

Inductive assume $\forall u$ $|u| = n$ for $|u| = n+1$
 let $u = aw$

$$\begin{aligned} (uv)^R &= (awv)^R \\ &= (wv)^R a \\ &= v^R w^R a \\ &= v^R u^R \end{aligned}$$

7 in the case of removing a different horse and adding the "H". it does not hold that the new group is all the same color

8 prove incorrect

$n^2 + n$ is odd for $n \geq 1$

theorem - base case

assume $n^2 + n$ is odd

$(n+1)^2 + (n+1)$ is also odd

$$(n+1)^2 + (n+1) = n^2 + 2n + 1 + n + 1 = (n^2 + n) + (2n + 2)$$

but $(n^2 + n)$ is odd by induction hypothesis... and

clearly $2n + 2$ is even. Thus $(n^2 + n) + (2n + 2)$

is the sum of an odd and even number
hence odd.

solution Base case of $n = 1$ does not hold.

Debate

- 1 4 hrs
- 2 moderate/hard
- 3 alone
- 4 45%
- 5